

Andrew Lau

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Education

University of California, Berkeley – Bachelor's degree, Computer Science; GPA: 3.8 August 2025 - Present

Expected Graduation: Spring 2028

Relevant Coursework: Data Structures and Algorithms, Discrete Math and Probability, Efficient Algorithms and Intractable Problems

Leadership: President of Generative AI @ Berkeley, first and largest student-run generative AI organization in California

Skills

Languages: Python, Java, Go, TypeScript, SQL, C, C++, Rust, Ruby

AI/ML: LLMs, RAG, Multimodal AI, Model Fine-Tuning, Prompt Compression, Context Optimization, Model Routing

Search & Data: Information Retrieval, BM25, Vector Search, Document Ingestion, Entity Extraction, Data Freshness

Experience

Pacific, Software Engineer Intern – San Francisco, CA June 2026 – Present

- Designed and built an **LLM optimization harness**, implementing **context compression**, **prompt caching**, and **model rerouting** to achieve **3x token reduction**.
- Built **enterprise search and freshness infrastructure**, implementing **ACL-aware document crawling**, **BM25 lexical retrieval**, **Qdrant sparse vectors**, and freshness workflows to optimize context and search, achieving **~45% prompt payload compression**.
- Engineered **document ingestion pipelines** using **DocETL** and **GLiNER**, fine-tuning models for **document understanding and entity extraction** while maintaining **>95% evaluation accuracy**.
- Architected a **distributed document lifecycle pipeline** with **idempotent execution** and state tracking, coordinating crawling, ingestion, reprocessing, and downstream synchronization at scale.

Apple, Machine Learning Engineer (Contract) – Berkeley, CA Feb 2026 – June 2026

- Architected and scaled a **distributed validation platform** for computer vision pipelines on cloud infrastructure, processing **10K+ images/day** with reliable, low-latency execution.
- Designed **LLM-driven, graph-based workflow orchestration** and **asynchronous job processing** systems backed by **message queuing**, enabling dependency-aware execution across complex validation pipelines.
- Led **end-to-end system design**, defining requirements and prioritizing features to translate ambiguous ML needs into **scalable platform capabilities**.

UC Berkeley EECS, Research Software Engineer – Berkeley, CA Sept 2025 – June 2026

- Owned end-to-end refactor and rebuild of a **30k+ LOC frontend in Next.js and React**, proactively identifying architectural gaps and cutting build and release timelines by **2+ months**.
- Optimized **CI/CD workflows** and deployment pipelines, reducing end-to-end build and deploy times by **~40%**
- Introduced **application-level caching** and state optimization (API response caching, memoization, and request deduplication), reducing redundant network calls by **~60%** and improving page load times by **~55%**.

Influcio, Software Engineer Intern – San Francisco, CA June 2025 – Sept 2025

- Led development of a **multimodal RAG pipeline**, coordinating cross-functional efforts and driving **system design decisions**.
- Trained a classifier that improved **data-collection accuracy by ~60%** across millions of items in large-scale automated workflows, increasing **evaluation throughput 1.5x**.

Projects

Orpheus Live

<https://github.com/andrewlau624/orpheus-live>

- Built a fully local voice-to-voice agent on Apple Silicon: mic, VAD, Whisper STT, Llama 3.2, and Orpheus TTS, with **barge-in** that cuts playback mid-sentence and an **LLM-driven turn-taking system** that dynamically chooses speak, wait, backchannel, or interrupt.